A good forecast can change with the evidence

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Revision / ES: revisión / PT: revisão

Name the change and show its effect.

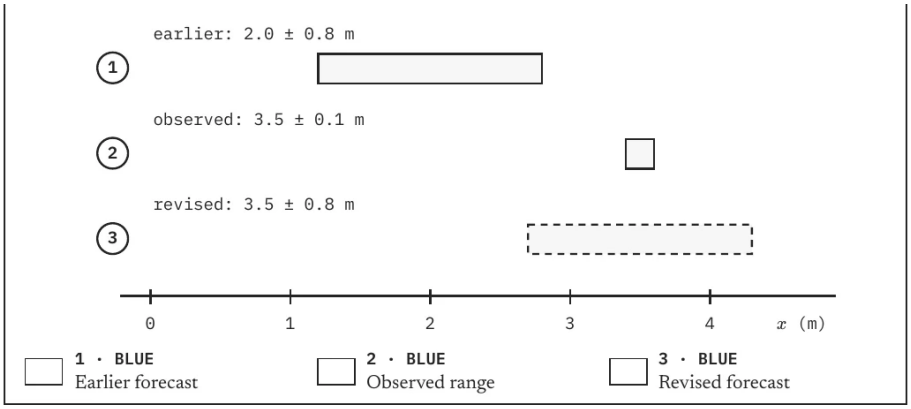
Also on your note page: Prediction, Evidence, Limit.

B READ THE RELATIONSHIP

$x_f = x_0 + vt$. With exact time, half-width = starting bound + t × velocity bound.

As you read, find: claim · evidence · limit.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

Keep $x_0 = 8 \pm 0.5$ m and exact $t = 3$ s . A new measurement changes v from −2.0 to −1.5 ± 0.1 m/s . Update the forecast.

Given $x_0 = 8 \pm 0.5$ m $v = -1.5 \pm 0.1$ m/s exact $t = 3$ s

Rule Update center and conservative bound.

Work $x_f = 8 - 1.5(3)$ $\delta x = .5 + 3(.1) = 0.8$ m

Answer 2.7 to 4.3 m Center shifts +1.5 m .

E SAVE FOR THE FINAL MEETING

Show the evidence behind XJ's forecast and what changed after review. Separate the physics from cost and feasibility decisions.

Read the physics: Revision

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Give a predicted position, a time and a reference frame. Show the measurement and calculation that support your answer. List any assumptions.

02 HOW IT WORKS

A model may give a central position and a range. A bound says which values the model allows; it does not give the chance of an impact. Compare the old and new results on one scale.

03 WHAT TO CHECK

Name the step that changed. Explain what it did to the forecast and what the physics still cannot decide.

04 TRY THE EXAMPLE FROM YOUR NOTES

Keep x 0 = 8 ± 0.5 m and exact t = 3 s . A new measurement changes v from −2.0 to −1.5 ± 0.1 m/s . Update the forecast. 2.7 to 4.3 m Center shifts +1.5 m .

Model check: $x_f = x_0 + vt$. With exact time, half-width = starting bound + t × velocity bound.

05 CONNECT TO XJ

Show the evidence behind XJ's forecast and what changed after review. Separate the physics from cost and feasibility decisions.

SAY IT BEFORE YOU WRITE

Which piece of evidence would make you revise again? We revised ___ after ___. Our forecast still assumes ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Prediction

ES: predicción / PT: previsão

Meaning in my words	How can you tell?
How would you explain prediction to a friend? In my words, this means ___.	Which part of a forecast could we check with the next camera image? I can tell because ___.
Example or sketch	**Non-example and why**
A tracker reads 0, 2, then 4 km a minute apart. Where would you look a minute later? My example shows ___.	Someone calls yesterday's recorded position a prediction. What would you call it? I would change ___ because ___.

Evidence

ES: evidencia / PT: evidência

Meaning in my words	How can you tell?
How would you explain evidence to a friend? In my words, this means ___.	What reading supports the claim that the cart reached 4 m in 2 s? I can tell because ___.
Example or sketch	**Non-example and why**
How could two measurements support a claim that one cart accelerated more? My example shows ___.	"I expected our cart to win." Does that tell us which cart crossed first? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Limit

ES: límite / PT: limite

Meaning in my words	How can you tell?
How would you explain limit to a friend? In my words, this means ___.	Why might a position forecast based on two blurry pictures have a wide range? I can tell because ___.
Example or sketch	**Non-example and why**
What limit could you mention when the team reports its best forecast? My example shows ___.	"Our forecast is definitely exact." Which detail about the camera would you ask for? I would change ___ because ___.

Revision

ES: revisión / PT: revisão

Meaning in my words	How can you tell?
How would you explain revision to a friend? In my words, this means ___.	Your predicted sighting is +6 km and the new image reads +5 km. What would you update? I can tell because ___.
Example or sketch	**Non-example and why**
What would you tell the next crew after that sighting? My example shows ___.	A team changes its answer only because another team disagrees. What would justify the change? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Doodle two forecast bands on the same scale. Add a note connecting the revision to a specific measurement.

Labels you can use: claim · evidence · limit.

Choose zero, direction, and scale.



Tell a partner

We revised ___ after ___. Our forecast still assumes ___.

Would you air this headline?

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SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Compare forecast ranges and keep conclusions within the data.

CHOOSE ONE JOB TOGETHER

FORECAST ENGINEER ☐

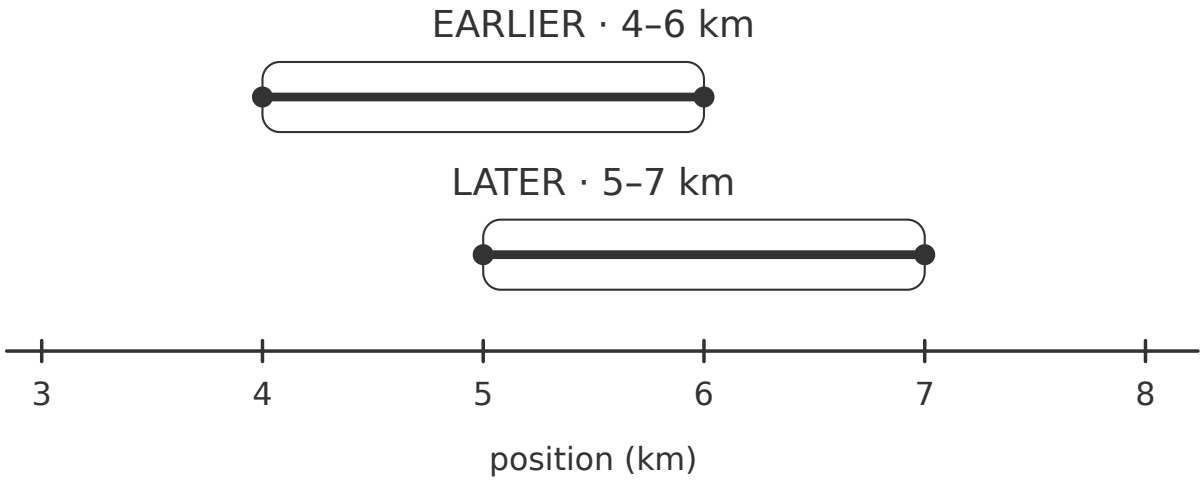
Mark both forecast bands. Issue a location bulletin and name an observation that would trigger another update.

SCIENCE NEWS EDITOR ☐

Decide whether to air the headline. Show both intervals and write the replacement line viewers should hear.

Word help: range / overlap / probability

Two short-interval forecasts in the same frame give **4-6 km** and **5-7 km**. The new estimate uses a later sighting. A headline says “Impact confirmed.”



01 OUR FIRST CALL

What location can you report? Should the proposed headline air? Explain.

The ranges show ___. They do not show ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Answer the caller using the bands. What can your desk report, and what is still unknown?

Now we would say ___ because ___.

Switch speakers. How is predicting XJ's location different from deciding what action people should take?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks